

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 748 Computational Photography — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
CMU 15-463 Computational Photography — Gkioulekas (free lectures, slides, assignments)	The primary source. Complete lecture notes and programming assignments covering most of this course, with unusually careful treatment of the physics.
Marc Levoy — Lectures on Digital Photography (free)	From the person who led Google's camera work and co-invented light field photography. Exceptional on optics, exposure, and why the phone pipeline looks the way it does.
Szeliski — Computer Vision: Algorithms and Applications, 2nd ed. (free online)	Chapters 3, 8, and 9 cover image processing, alignment, and stitching. The standard reference, free, and the alignment material for Module 07 is the best there is.
Georgia Tech CS6475 Computational Photography (free)	A second voice with more worked examples and a gentler pace. Good for Modules 03, 04, and 08.
Reinhard et al. — High Dynamic Range Imaging	The standard HDR reference. Library copy; the key papers by the same authors are free and linked per module.
Hasinoff et al. — Burst photography for high dynamic range and low-light imaging (free)	The HDR+ paper, with a full dataset of raw bursts. This is what your phone does, documented by the people who built it.

Module-by-module map

Module 01 — What a Camera Measures

Resource	What it covers for this module
Marc Levoy — Lectures on Digital Photography, lectures 1–6 (free)	Image formation, exposure, and sensors, from someone who built camera pipelines. The clearest treatment of §1 available free.
CMU 15-463 — Image formation and the camera pipeline (free)	The lecture notes and the raw-processing assignment, which is essentially Project 1's first requirement.
Emil Martinec — Noise, Dynamic Range and Bit Depth in Digital SLRs (free)	The clearest explanation of §3 for photographers, with measurements. The site also publishes measured sensor data.
Hasinoff — Photon, Poisson Noise (free chapter)	The noise model of §3 stated precisely, by one of the HDR+ authors.

Module 02 — Image Processing Foundations

Resource	What it covers for this module
Szeliski — Computer Vision, Chapter 3 (free)	Linear filtering, pyramids, and edge-aware filters, with the frequency analysis. The reference for this module.
CMU 15-463 — image processing and pyramids lectures (free)	Convolution, pyramids, and the assignment that builds them.
Paris et al. — Bilateral Filtering: Theory and Applications (free)	The definitive survey of §3, including the fast approximations and the joint bilateral applications.
He, Sun & Tang — Guided Image Filtering (free)	The guided filter. Short paper, linear time, and a few lines of code.

Module 03 — High Dynamic Range

Resource	What it covers for this module
Debevec & Malik — Recovering High Dynamic Range Radiance Maps from Photographs (1997, free)	The paper. The log-linearisation of §2 and the weighting scheme, in the original, and still clearly written.
Hasinoff et al. — Burst photography for HDR and low-light imaging (free, with dataset)	The §4 approach, documented by the team that shipped it, with a dataset of raw bursts to work on.
CMU 15-463 — HDR imaging lecture and assignment (free)	The merging assignment, which is essentially Project 1's HDR requirement.
Tursun et al. — The State of the Art in HDR Deghosting (free)	A survey of §3's techniques, with comparisons.

Module 04 — Tone Mapping

Resource	What it covers for this module
Reinhard et al. — Photographic Tone Reproduction for Digital Images (free)	The global operator of §2, including the photographic analogy that motivates it.
Durand & Dorsey — Fast Bilateral Filtering for the Display of HDR Images (free)	The bilateral tone mapping of §3, and one of the papers that brought the bilateral filter into graphics.
Fattal, Lischinski & Werman — Gradient Domain High Dynamic Range Compression (free)	The gradient-domain method, and a natural lead-in to Module 08.
Mantiuk et al. — tone mapping evaluation work (free)	Perceptual evaluation of tone mapping, including the accurate-versus-preferred distinction in §4.

Module 05 — Demosaicing and the Raw Pipeline

Resource	What it covers for this module
CMU 15-463 — the image processing pipeline lecture and assignment (free)	The complete raw pipeline, implemented. This assignment is Project 1's first requirement.
Malvar, He & Cutler — High-quality linear interpolation for demosaicing (free)	The gradient-corrected method of §1. A fixed linear filter, a few lines, and close to state of the art for a long time.
Gijzenij, Gevers & van de Weijer — Computational Color Constancy survey (free)	The illuminant estimation methods of §2, compared on standard datasets.
Adobe DNG specification (free)	What is actually in a raw file, including the colour matrices and how they are intended to be applied.

Module 06 — Deconvolution and Deblurring

Resource	What it covers for this module
CMU 15-463 — deconvolution and coded photography lectures (free)	Wiener, Richardson–Lucy, and the coded aperture material, with the assignment.
Fergus et al. — Removing Camera Shake from a Single Photograph (free)	The §3 breakthrough. The natural image prior argument is the part to read closely.
Levin et al. — Image and Depth from a Conventional Camera with a Coded Aperture (free)	The coded aperture of §4, including the depth recovery.
Raskar, Agrawal & Tumblin — Coded Exposure Photography: Motion Deblurring using Fluttered Shutter (free)	The flutter shutter. The spectral argument is clearly made.

Module 07 — Alignment and Stitching

Resource	What it covers for this module
Szeliski — Computer Vision, Chapters 8 and 9 (free)	Alignment, motion models, RANSAC, bundle adjustment, and stitching. The reference for this module , and Szeliski wrote the standard panorama survey as well.
Lowe — Distinctive Image Features from Scale-Invariant Keypoints (free)	SIFT, including the ratio test of §2. One of the most cited papers in computer vision, and readable.
Brown & Lowe — Automatic Panoramic Image Stitching (free)	The complete pipeline of this module, end to end, including bundle adjustment and multi-band blending.
Burt & Adelson — A Multiresolution Spline with Application to Image Mosaics (free)	Multi-band blending, from 1983 and still the method in use.

Module 08 — Gradient-Domain Processing

Resource	What it covers for this module
Pérez, Gangnet & Blake — Poisson Image Editing (free)	The paper. Seamless cloning, mixed gradients, and the variants of §3, in eight pages. One of the most immediately useful graphics papers ever written.
CMU 15-463 — gradient-domain processing lecture and assignment (free)	The derivation and the implementation. The assignment is Project 2's third option.
Bhat et al. — GradientShop: A Gradient-Domain Optimization Framework (free)	The general framework of §3 — many image operations as one gradient-domain optimisation.
Farbman et al. — Coordinates for Instant Image Cloning (free)	Mean-value coordinates — the fast approximation in §4.

Module 09 — Light Fields and Plenoptic Cameras

Resource	What it covers for this module
Levoy & Hanrahan — Light Field Rendering (1996, free)	The paper that introduced the 4D parameterisation of §1. Short and foundational.
Ng — Digital Light Field Photography (thesis, free)	The plenoptic camera and the refocusing mathematics of §3, by Lytro's founder. The Fourier slice refocusing is here.
Stanford Light Field Archive (free data)	Captured light fields to experiment with, including the camera array datasets. You can implement refocusing on real data without any hardware.
Mildenhall et al. — NeRF (free)	The §4 connection. Read it as a light field paper and the lineage is obvious.

Module 10 — Burst Photography

Resource	What it covers for this module
Hasinoff et al. — Burst photography for high dynamic range and low-light imaging (free, with dataset)	The HDR+ paper and the raw burst dataset. This module is essentially a reading of it, and the dataset lets you implement the pipeline on real data.
Wronski et al. — Handheld Multi-Frame Super-Resolution (free)	The §4 result, including the deliberate motion on a tripod.
Levoy & Pritch — Night Sight blog posts (free)	How the pipeline was extended to very low light, written by the people who did it.
Liba et al. — Handheld Mobile Photography in Very Low Light (free)	Motion metering, exposure selection, and the limits in §4.

Module 11 — Depth, Matting, and Synthetic Defocus

Resource	What it covers for this module
Wadhwa et al. — Synthetic Depth-of-Field with a Single-Camera Mobile Phone (free)	The portrait mode paper. Dual-pixel depth, segmentation, and the defocus rendering of §3, from the team that shipped it.
Levin, Lischinski & Weiss — A Closed-Form Solution to Natural Image Matting (free)	The matting Laplacian of §2. Elegant, and a third appearance of the same linear system.
Chuang et al. — A Bayesian Approach to Digital Matting (free)	The Bayesian formulation, and a clear statement of the underdetermination.
Szeliski — Computer Vision, Chapter 12 (free)	Stereo and depth estimation, with the failure analysis of §1.

Module 12 — Learned Image Processing

Resource	What it covers for this module
Blau & Michaeli — The Perception-Distortion Tradeoff (free)	The theorem in §2. Read it before making any claim that one method is better than another.
Zhang et al. — The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS, free)	Perceptual metrics, and a clear demonstration of where PSNR fails to track human judgement.
Chen et al. — Learning to See in the Dark (free)	End-to-end raw-to-RGB in extreme low light. The clearest demonstration of the §1 argument, with a dataset.
Menon et al. — PULSE, and the subsequent discussion (free)	The upscaling case of §3, including the bias findings that followed publication. Read the paper and the critique together.

Module 13 — What a Photograph Is Now

Resource	What it covers for this module
C2PA — Coalition for Content Provenance and Authenticity (free specification)	The provenance standard of §2. The specification is readable and the threat model section is the honest part.
Farid — Photo Forensics	The forensic analysis of §2, from the leading researcher. His papers and course material are free.
Levoy — writings on computational photography and what it means (free)	Thoughtful on the §1 question from someone who built the pipelines that raised it.
Reuters and AP photo manipulation standards (free)	How news organisations actually draw the line — worth reading because they had to be specific where academics can be vague.